

Himanshu Khichi

📍 Bengaluru, India | 📞 +91 7742435717 | ✉ himanshukhichi99@gmail.com | 🌐 himanshukhichi.vercel.app | 📱 himanshukhichi

SUMMARY

Backend Software Engineer with **5 years** of experience at Oracle building core platform features for MMG — an enterprise ML model lifecycle and governance platform for financial institutions, spanning model development, deployment, and regulatory compliance. Shipped a multi-tenant model serving platform, 99.9% storage-reducing statistical dataset profiling, data drift detection service, and led full SaaS migration to cloud (OCI). Built a distributed key-value store sustaining **50K+ ops/sec** and a DAG job scheduler handling **1,000+ concurrent tasks** with Kafka exactly-once delivery and Redis leader election.

WORK EXPERIENCE

Oracle (OFSS)

Bengaluru, India

Software Engineer — ML Platform, Model Management & Governance

Aug 2024 – Present

- Architected a **multi-tenant ML model serving platform** exposing models as authenticated REST APIs with multi-model hosting, per-request audit logging, and Swagger UI testing.
- Led **end-to-end SaaS migration** of the ML platform (9 microservices) from on-premises to cloud infrastructure (OCI); implemented a unified codebase strategy maintaining consistency across on-prem and SaaS environments and built DevOps **CI/CD pipelines** automating Docker builds and Artifactory publishing across all microservices.
- Developed **conda management APIs** to provision isolated conda environments from Object Storage specifications, eliminating manual admin workflows (inline Python scripts) and reducing setup time by **~80%**.

Associate Software Engineer — ML Platform, Model Management & Governance

Jul 2021 – Aug 2024

- Reduced per-dataset storage by **99.9%** (GBs → KBs) by building a **scheduled profiling pipeline** generating compact binary profiles capturing per-feature statistics across thousands of datasets.
- Designed a **data drift detection service** over dataset profiles — computes per-feature drift between baseline and target windows against configurable thresholds, triggering alerts for proactive model degradation detection.
- Built **dataset profiling analytics** — per-feature data quality stats, correlation matrices, and cross-profile comparisons, enabling data scientists to track dataset evolution across ingestion windows and catch data quality regressions early.

PROJECTS

DistKV — Distributed Key-Value Store | Java, gRPC, Protobuf, Docker, Prometheus, Grafana

- Built a distributed KV store with **consistent hashing** (150 virtual nodes), tunable **quorum replication** (N/R/W configurable), gossip-based failure detection, hinted handoff, and vector clocks — sustaining **50K+ ops/sec** at sub-10ms P99 latency across a 5-node cluster.
- Per-node storage engine with **write-ahead log** for crash recovery, LRU eviction, and **bloom filters** cutting unnecessary lookups by **~90%**; gRPC API with streaming Scan RPC and Prometheus + Grafana observability.

FlowMesh — Distributed DAG Job Scheduler | Java, Kafka, Redis, PostgreSQL, gRPC, Docker

- Distributed DAG scheduler with **Redis Redlock leader election**, Kafka task distribution, and **exactly-once execution** via transactional producer + PostgreSQL deduplication — scheduling **1,000+ concurrent tasks** under 500ms P99 latency.
- **DFS cycle detection**, Kahn's topological sort, 6-state task machine with exponential backoff retry and **dead-letter queue** recovery; gRPC worker heartbeats for automatic failure detection and task re-queuing.

SKILLS

Languages: Java, C/C++, Python, SQL, JavaScript, Protocol Buffers

Technologies: Spring Boot, gRPC, Apache Kafka, Redis, PostgreSQL, REST APIs, Microservices, Docker, Kubernetes, AWS, OCI, CI/CD, Prometheus, Grafana, JUnit 5, Mockito

Concepts: Distributed Systems, Concurrency, Consistent Hashing, Quorum Replication, Leader Election, Exactly-Once Semantics, Event-Driven Architecture, Observability, System Design, Agile/Scrum, Code Review

EDUCATION

National Institute of Technology Warangal (NIT Warangal)

Warangal, India

B.Tech, Computer Science & Engineering

Jul 2017 – Jun 2021

- Coursework: Data Structures & Algorithms, Distributed Systems, Operating Systems, Database Systems, Computer Networks